Free-body diagrams

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Draw only the external forces on one chosen object.

SUCCESS CRITERIA - I can...

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An a-versus-F_{net} graph has slope **1.5** $(m/s^2)/N$.
1. **Find mass** and acceleration at **2.5 N**.
2. **Explain why a driving-force plot** can have an offset.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = **9.8 N/kg** unless another value is given.

Worked reasoning: A lamp hangs motionless from one vertical cable. With weight **12 N** down, the cable tension is **12 N** up. Only forces acting on the lamp belong in its diagram.

Your turn: 1. **Draw the lamp as a** labeled point with its two force arrows.
2. **Explain why its force** on the cable is not another arrow on this diagram.

Add and label force arrows.

Lamp: draw the two 12 N forces acting on it.

Marzano check - circle: ① ② ③

03 Motionless lamp

A **2 kg** lamp hangs at rest from one vertical rope.
1. Use g = **9.8 N/kg**.
2. Draw and **label the forces** on the lamp, including their magnitudes.

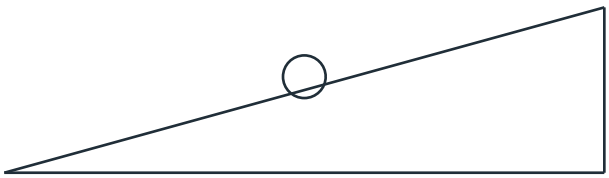
Add and label force arrows.

The dot represents the object. Object: 2 kg lamp

Marzano check - circle: ① ② ③

04 A rough ramp

A block rests on a rough incline.
1. **Add weight**, normal force and static friction to the ramp diagram.
2. **Explain why weight points vertically** downward rather than perpendicular to the surface.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Critique the extra arrows

A student's free-body diagram includes weight, normal force, friction, a "velocity force," and an extra "net force." Redraw a correct diagram for a box sliding right on a rough level floor after release. **Explain each removal**.

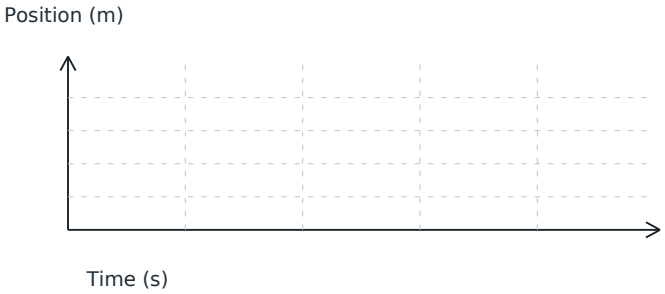
Add and label force arrows.

Box after release; it slides right.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A straight position-time segment goes from (**2 s**, **-4 m**) to (**6 s**, **20 m**). **Find velocity**, interpret its units and **predict x** at **8 s** if motion continues.
Label the diagram with the given positions and any prediction.



Marzano check - circle: 1 2 3

07 Exit check

A box slides to the left on a rough level floor after release. **Draw all forces**; **state what can be concluded** about vertical and horizontal acceleration.

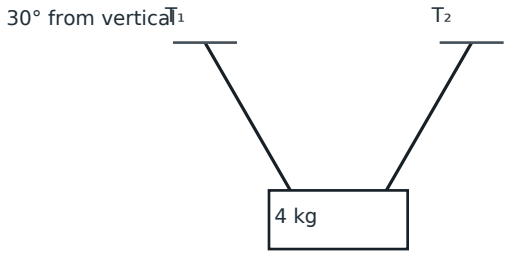
Add and label force arrows.

Box slides left after release on a rough level floor.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two identical cables support a stationary **4 kg** object. Each cable is **30°** from vertical.
1. Use g = **10 N/kg**.
2. **Add force arrows** to the diagram and **calculate the tension** in each cable.



Marzano check - circle: 1 2 3